AI FAQ Assistant API

Instant, intelligent answers to frequently asked questions

Presented by: Your Name | Department / College | Date

Introduction

- Users ask the same questions again and again on websites, apps and support desks
- Manual replies are slow, costly and not available 24/7
- AI can understand a question and reply instantly with the right answer
- The AI FAQ Assistant API offers this ability as a simple service any app can plug into

What is an AI FAQ Assistant?

- An AI-powered system that answers user questions automatically from a FAQ knowledge base
- Uses Natural Language Processing (NLP) to understand the meaning of a question, not just keywords
- Works even when the question is phrased differently from the stored FAQ
- Exposed as an API, so websites, mobile apps and chatbots can use it easily

Need for the System

24/7 Availability

Answers are available any time without human agents

Faster Response

Replies in seconds instead of hours or days

Reduced Workload

Support staff handle only complex issues

Consistent Answers

Every user gets the same accurate information

Objectives

- Build an API that answers user questions automatically
- Understand natural language queries using AI
- Return accurate, relevant answers from the FAQ data
- Provide fast response with simple integration for any application
- Make the FAQ content easy to update and extend

How the API Works

1. User asks

The user types a question in the app or website

2. API receives

The question is sent to the API as a request

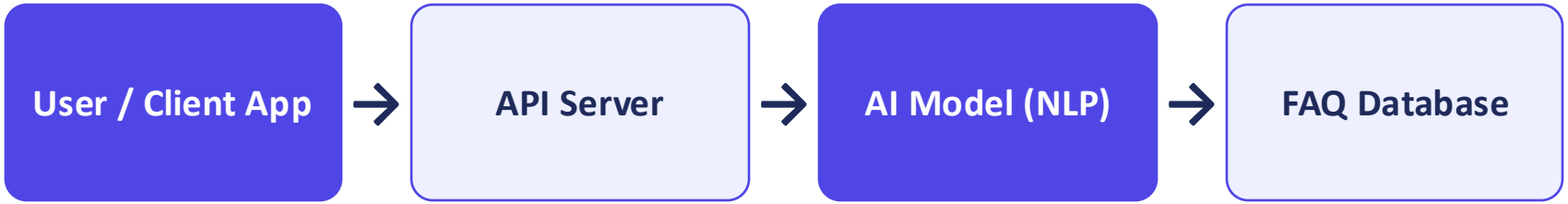
3. AI processes

The AI matches the question with the best FAQ answer

4. Answer returned

The API sends the answer back as a response

System Architecture



Request flows left to right; the answer travels back to the user as a response.

Technologies Used

Python

Core programming language

FastAPI / Flask

Builds the REST API

NLP / AI Model

Understands questions and finds answers

JSON

Request and response data format

Database

Stores FAQ questions and answers

Postman

Testing the API endpoints

API Request & Response

Request (POST /ask)

```
{
  "question": "How do I
  reset my password?"
}
```

Response

```
{
  "answer": "Click Forgot
  Password on the login
  page and follow the
  steps.",
  "confidence": 0.94
}
```

Advantages

- Instant responses, available around the clock
- Saves time and support cost
- Easy to integrate with any website or app
- Understands different ways of asking the same question
- Scalable: handles many users at the same time

Applications

Customer Support

Automated help desks and chatbots

Education

Student queries about courses and admissions

E-commerce

Orders, delivery and return questions

Banking & Healthcare

Common service and policy questions

Conclusion

- The AI FAQ Assistant API automates answering repeated questions
- It improves speed, availability and user satisfaction
- Simple API design makes it easy to plug into any application
- Future scope: multi-language support, voice input and learning from user feedback

Thank you!